



PREDICTIVE DATA ANALYTICS

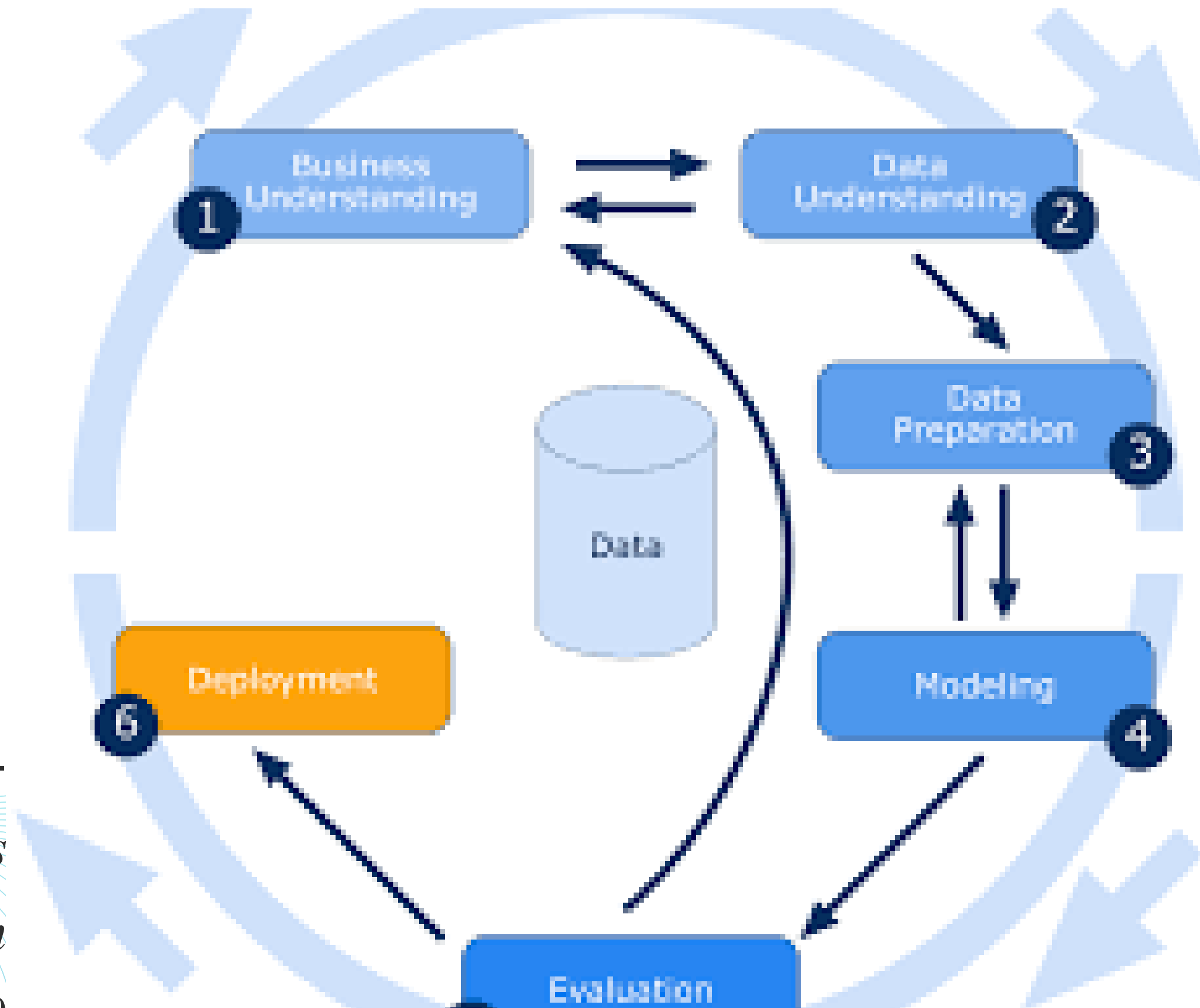
LESSON 2: DATA EXPLORATION & DATA PREPARATION

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CONTENT

- **Problem identification**
- **Data exploration**
- **Data Preparation**

CRISP-DM



The CRISP-DM (Cross-Industry Standard Process for Data Mining), Wirth and Hipp 2000

1. PROBLEM IDENTIFICATION

Guides:

- *What is the business problem? What are the goals that the business wants to achieve?*
- *How does the business currently work?*
- *In what ways could a predictive analytics model help to address the business problem?*

Case study : Vehicle insurance fraud

- **Business problem:** in spite of having a fraud investigation team that investigates up to 30% of all claims made, a vehicle insurance company is still losing too much money due to fraudulent claims.
- **Predictive analytics solutions:** **Claim prediction** (the likelihood that an insurance claim is fraudulent), **Member prediction** (the propensity of a member to commit fraud in the near future), **Application prediction** (the likelihood that a policy someone has applied for will ultimately result in a fraudulent claim), **Payment prediction** (the amount most likely to be paid out by an insurance company after having investigated a claim.)

2. DATA EXPLORATION



Vehicle Insurance Claim Fraud Detection

Fraud detection use-case for vehicle insurance industry

[kaggle.com](https://www.kaggle.com)

- 33 features, 15420 rows
- [Code example](#)

01) Definition of Features

[25]:

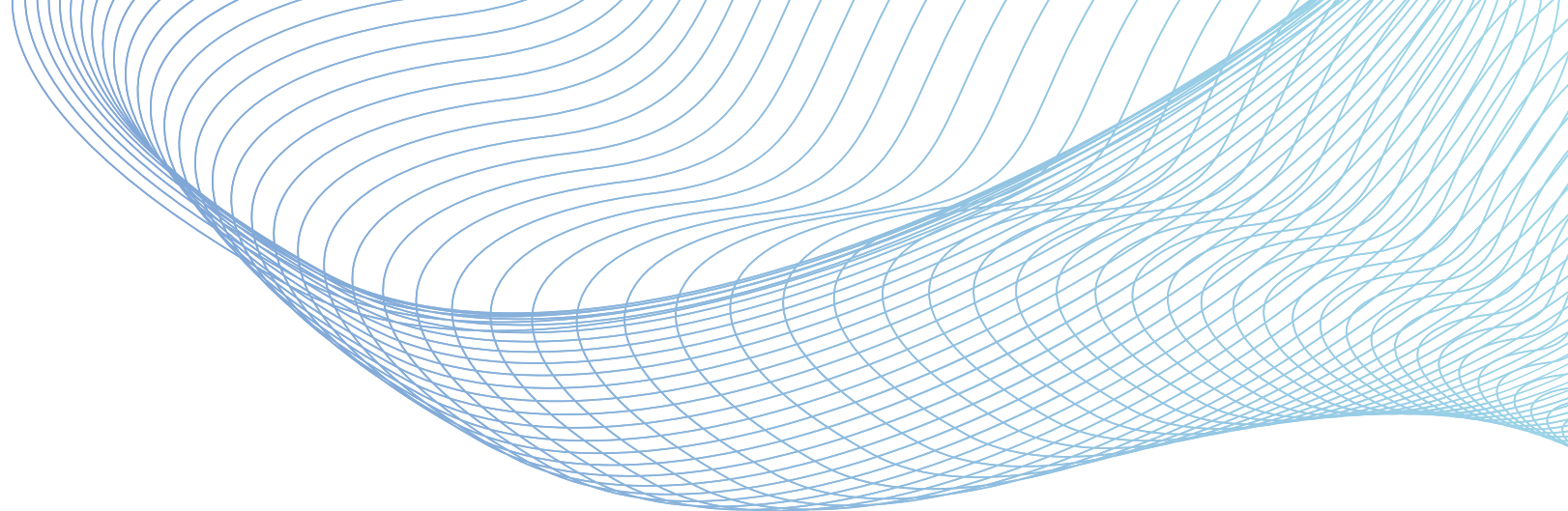
```
raw_data.head()
```



[25]:

	Month	WeekOfMonth	DayOfWeek	Make	AccidentArea	DayOfWeekClaimed	MonthClaimed	WeekOfMonthClaimed	Sex	MaritalStatus	Age	Fault
0	Dec	5	Wednesday	Honda	Urban	Tuesday	Jan	1	Female	Single	21	Policy Holder
1	Jan	3	Wednesday	Honda	Urban	Monday	Jan	4	Male	Single	34	Policy Holder
2	Oct	5	Friday	Honda	Urban	Thursday	Nov	2	Male	Married	47	Policy Holder

DATA TYPES



Continuous data types:

- **Numeric:** True numeric values that allow arithmetic operations (e.g., price, age)
- **Interval:** Values that allow ordering and subtraction, but do not allow other arithmetic operations (e.g., date, time)

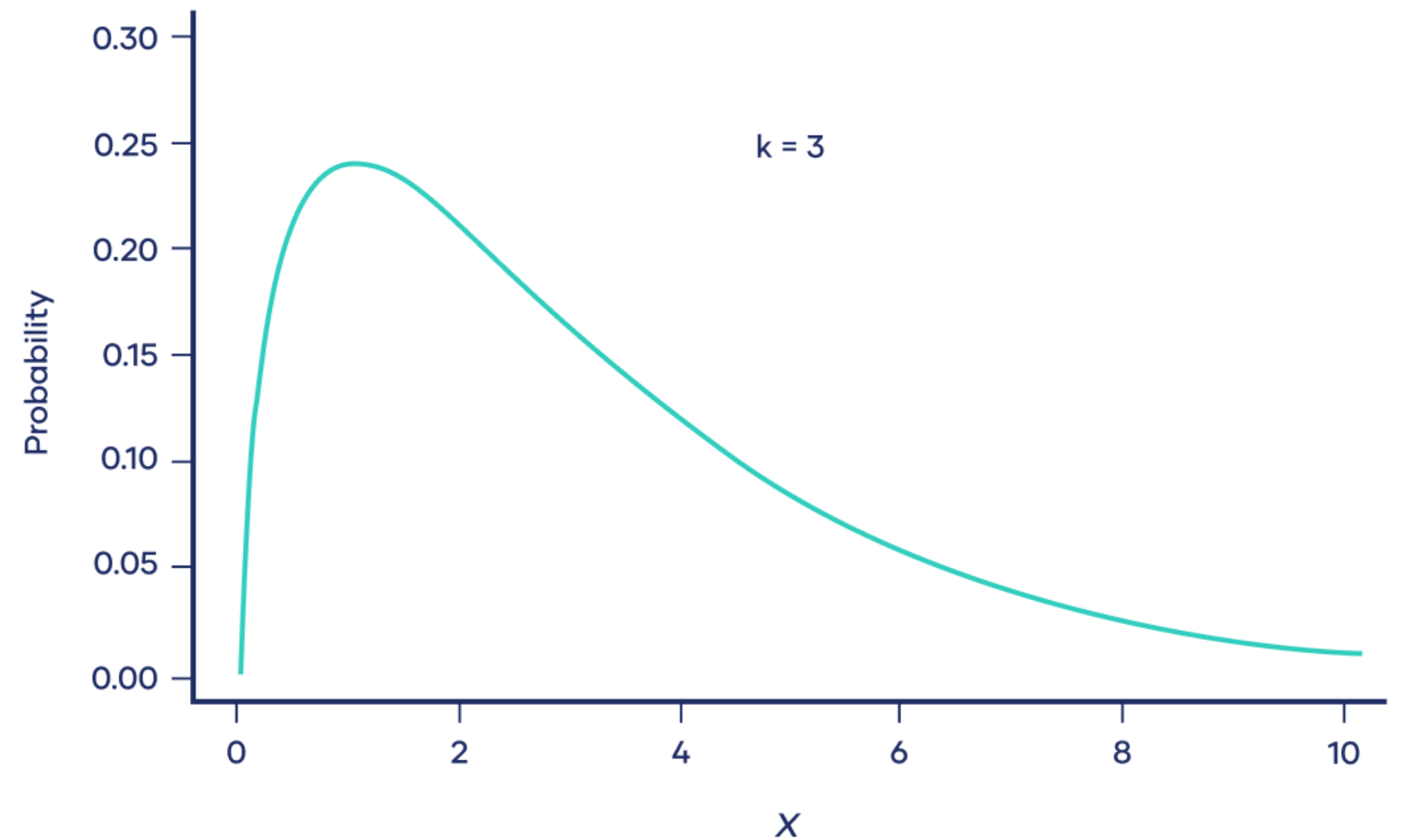
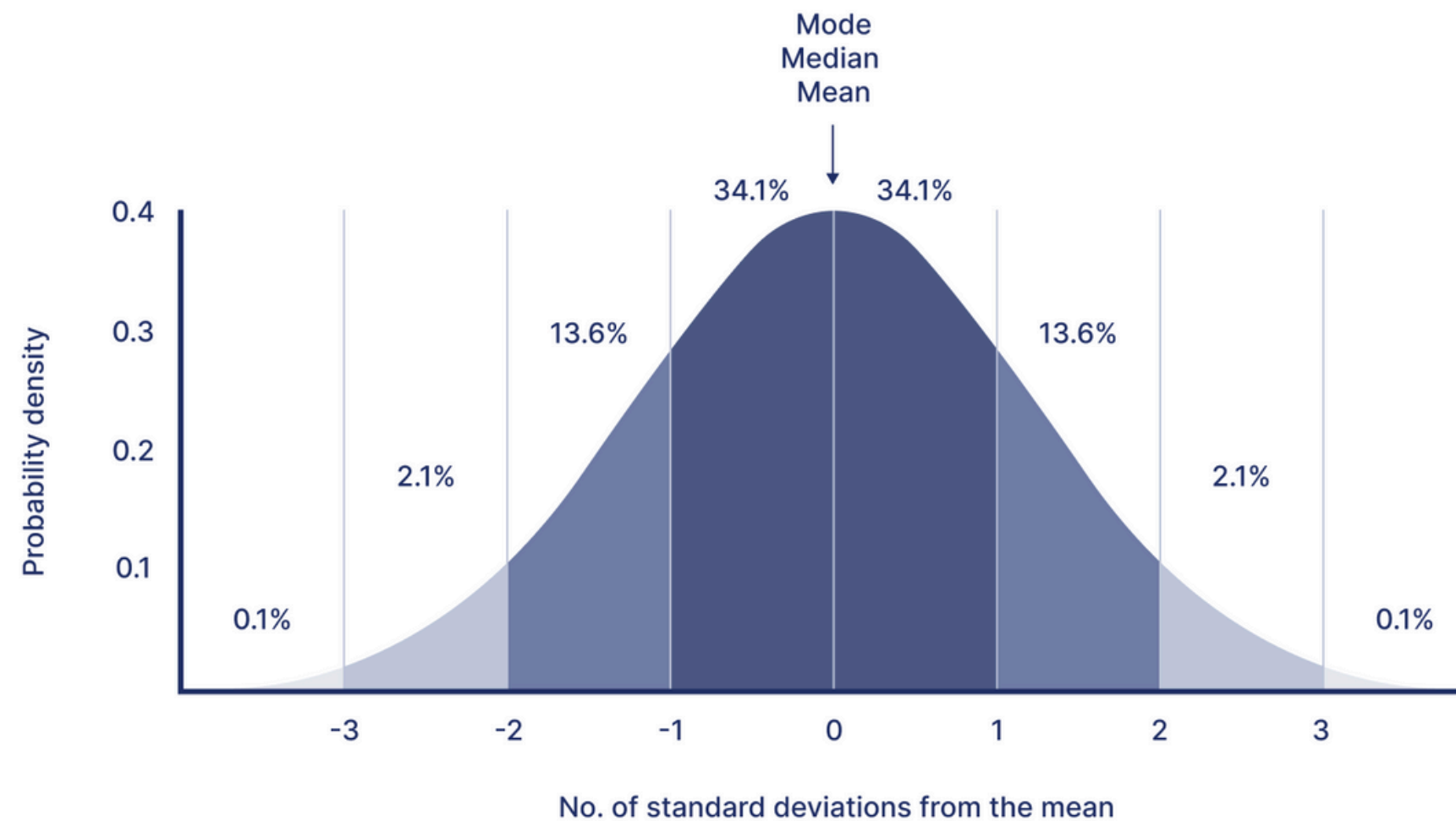
Categorical data types:

- **Ordinal:** Values that allow ordering but do not permit arithmetic (e.g., size measured as small, medium, or large)
- **Categorical:** A finite set of values that cannot be ordered and allow no arithmetic (e.g., country, product type)
- **Binary:** A set of just two values (e.g., gender)
- **Textual:** Free-form, usually short, text data (e.g., name, address)

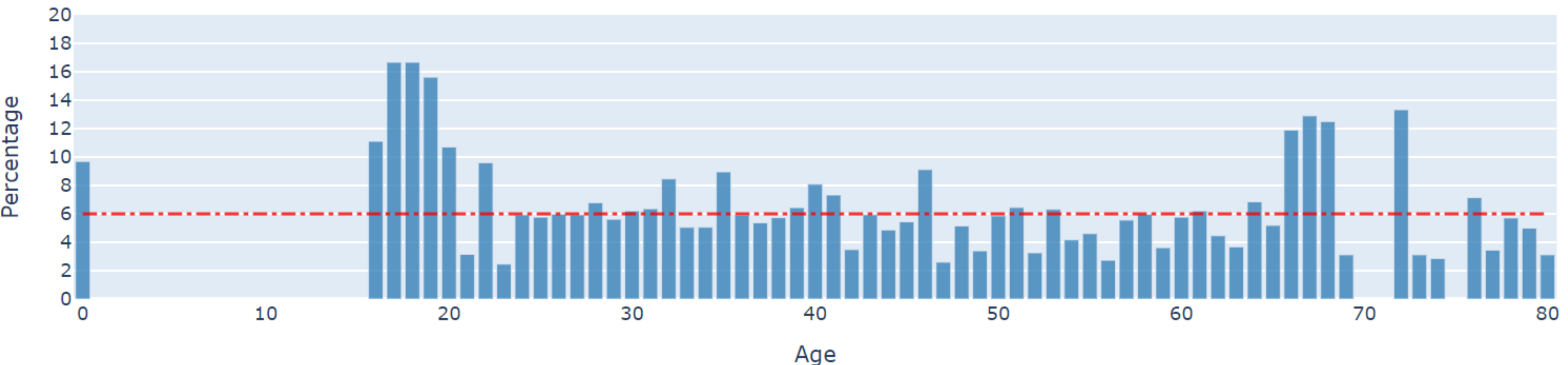
DATA DISTRIBUTIONS

Chi-Square Distribution

Standard normal distribution

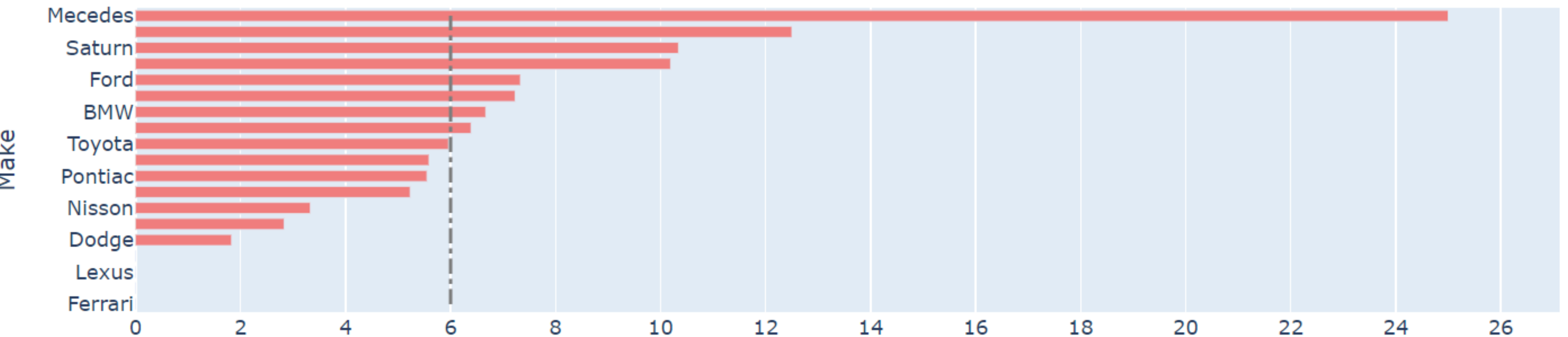


Fraud Detection by Age



- Fraud is most frequently detected among teenagers and retired seniors.
- Teenagers have weak financial power as they have not yet started their economic activities, and retired seniors experience weakened financial power post-retirement.
- There seems to be a correlation between financial power and fraud

Fraud Detection by Make



- The Mercedes owners are more likely to be involved in fraud, with almost double the incidence compared to the second-highest group, Accura owners.
- On the other hand, the Porche, Lexus, Jaguar, Ferrari owners have never been reported for fraud; all the four make are "High-End"

3. DATA PREPARATION

- **Data Cleaning:** Handling Missing Values, Removing Duplicates, Correcting Inaccurate Data, Ensuring Consistency, Handling Outliers
- **Data Integration:** Combining data from different sources into a coherent dataset.
- **Data Transformation:** Normalization and Scaling, Encoding Categorical Variable, Feature Engineering, Aggregation
- **Data Reduction:** Dimensionality Reduction, Feature Selection, Sampling
- **Data Discretization:** Converting continuous data into discrete buckets
- **Data Splitting:** Dividing data into training, validation, and test sets.

ENCODING CATEGORICAL VARIABLES

Label Encoding: transform non-numerical labels into numerical labels

'Female': 0, 'Male': 1

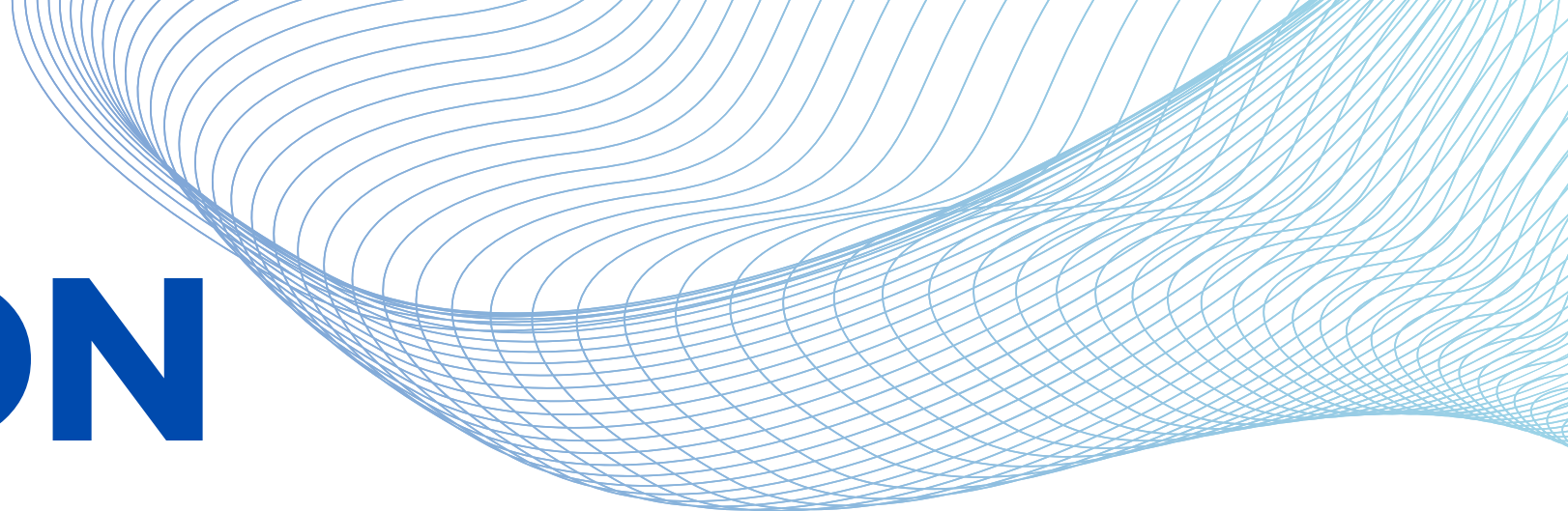
An Ordinal Categorical Variable is a categorical variable where the categories have a logical order or ranking (e.g. low: 0, medium: 1, high: 2)

One Hot Encoding

- convert categorical data into a binary (0 or 1). Each category is represented by a binary vector with the length equal to the number of categories. In this vector, all elements are 0 except for the element corresponding to the particular category, which is set to 1.

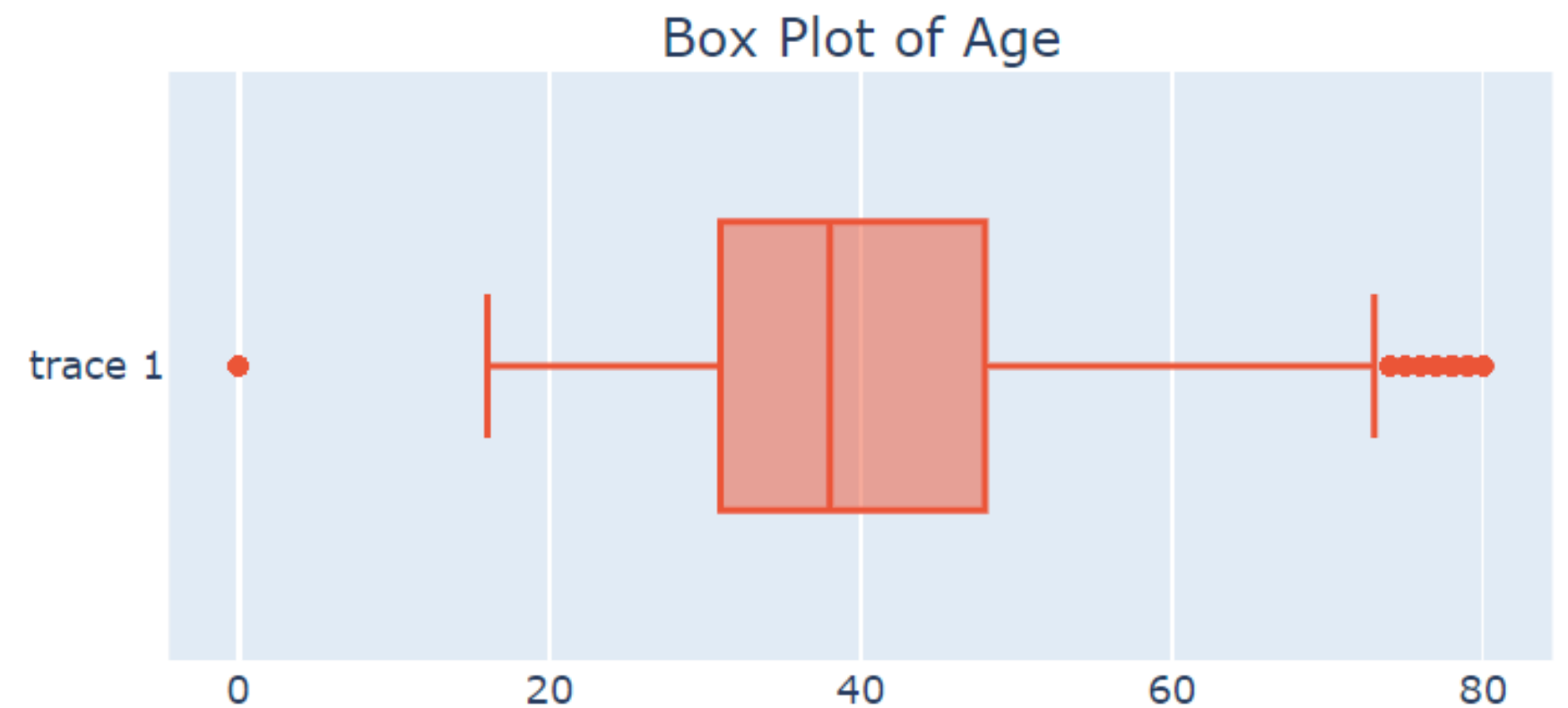
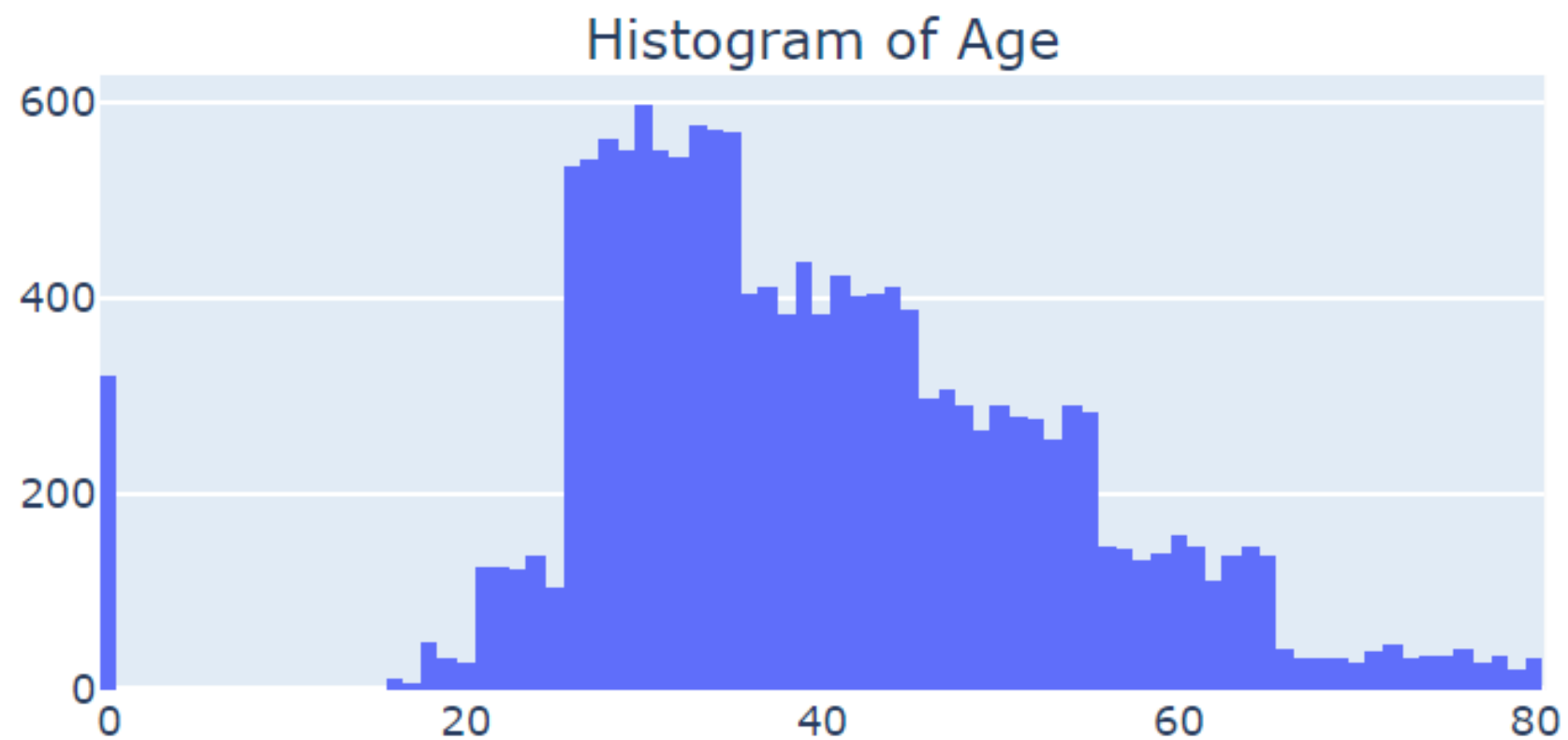
'Red' -> [1, 0, 0] *'Green' -> [0, 1, 0]* *'Blue' -> [0, 0, 1]*

DATA REDUCTION



- Drop Useless Columns
- Drop Constant Features

DATA CLEANING



- Correcting Inaccurate Data
- Handling Outliers: Outlier Replacement using LightGBM

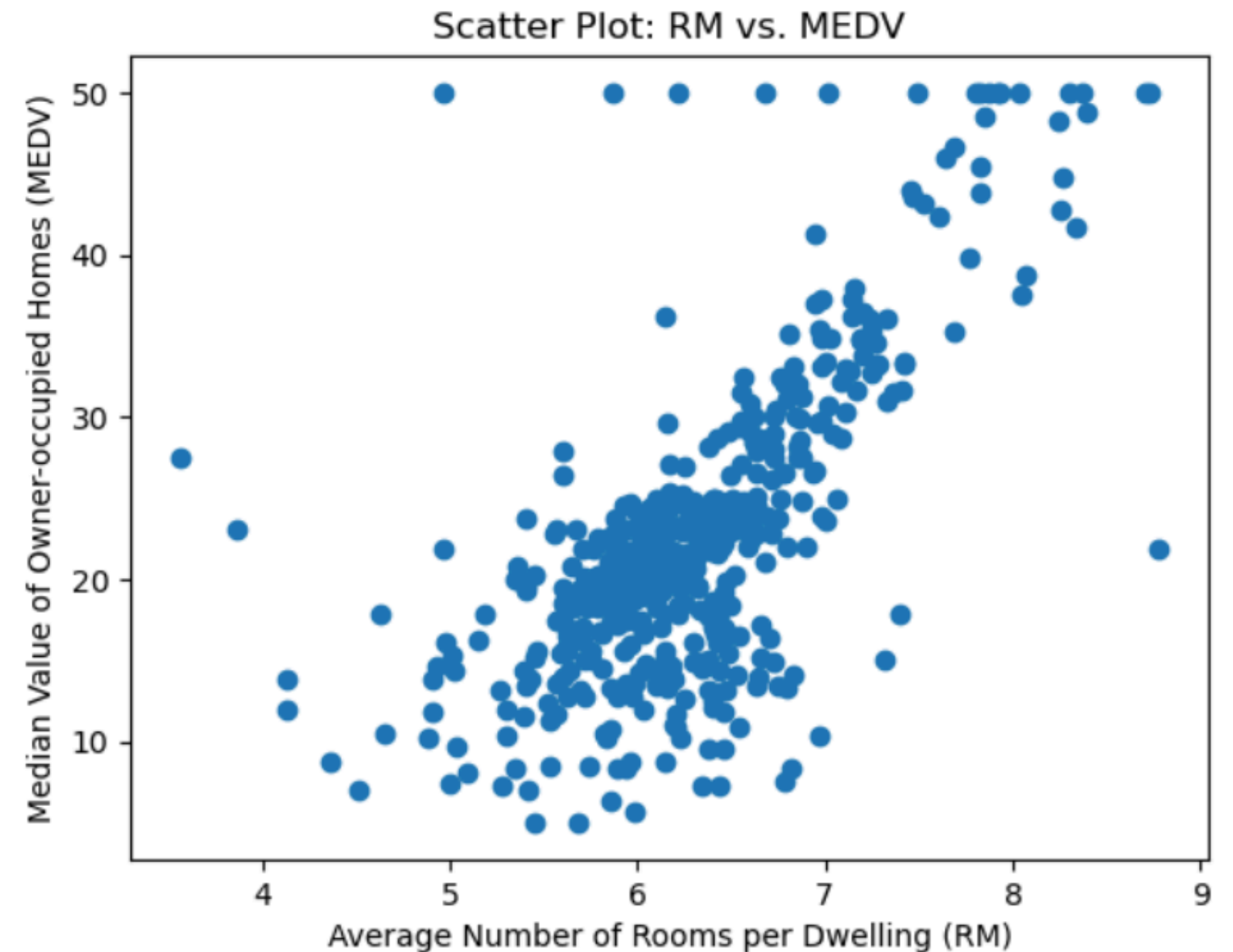
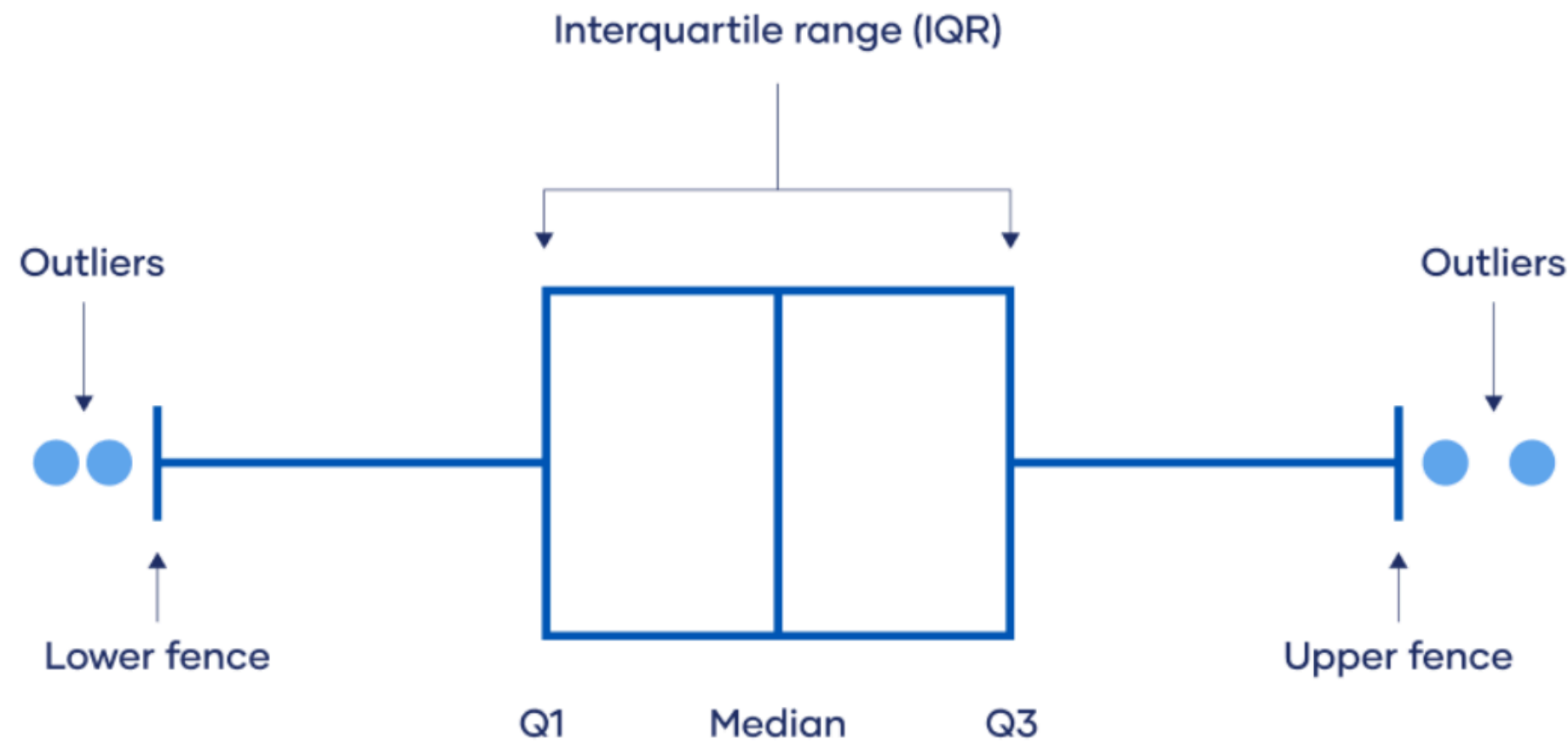
OUTLIER DETECTION METHOD

Statistical Technique:

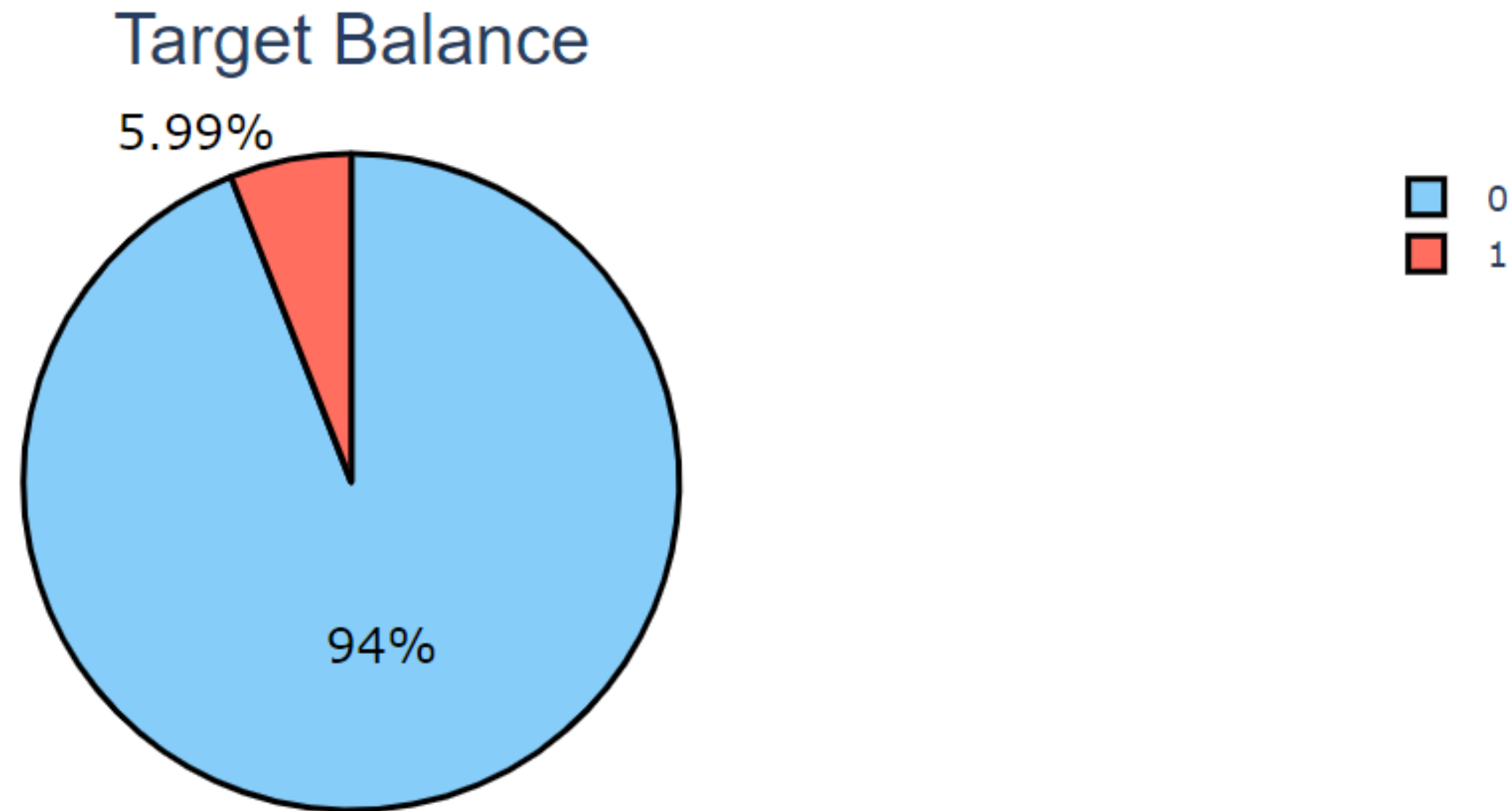
- *z-score* measures the number of standard deviations a data point is away from the mean of a distribution. It identifies outliers as data points that fall outside a specified threshold (typically, a z-score threshold of 2 or 3)
- The *IQR* method involves calculating the spread of the data by determining the range between the first quartile (25th percentile) and the third quartile (75th percentile) of a distribution. Outliers are identified as data points that fall below the first quartile minus a specified threshold or above the third quartile plus the threshold

OUTLIER DETECTION METHOD

Visualization technique



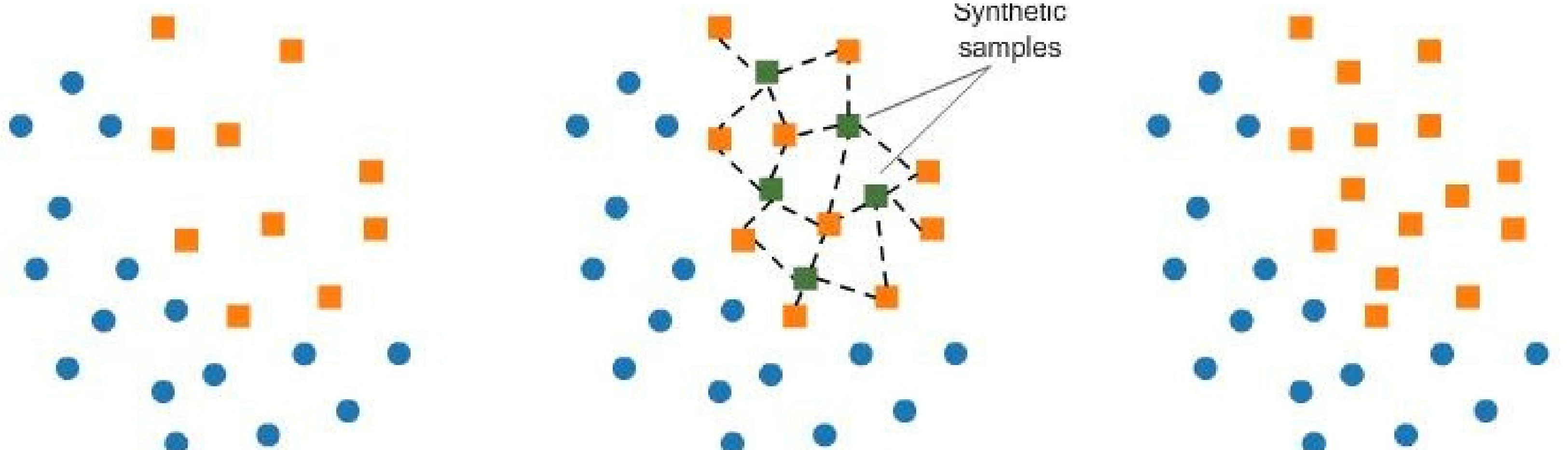
HANDLING IMBALANCE DATA



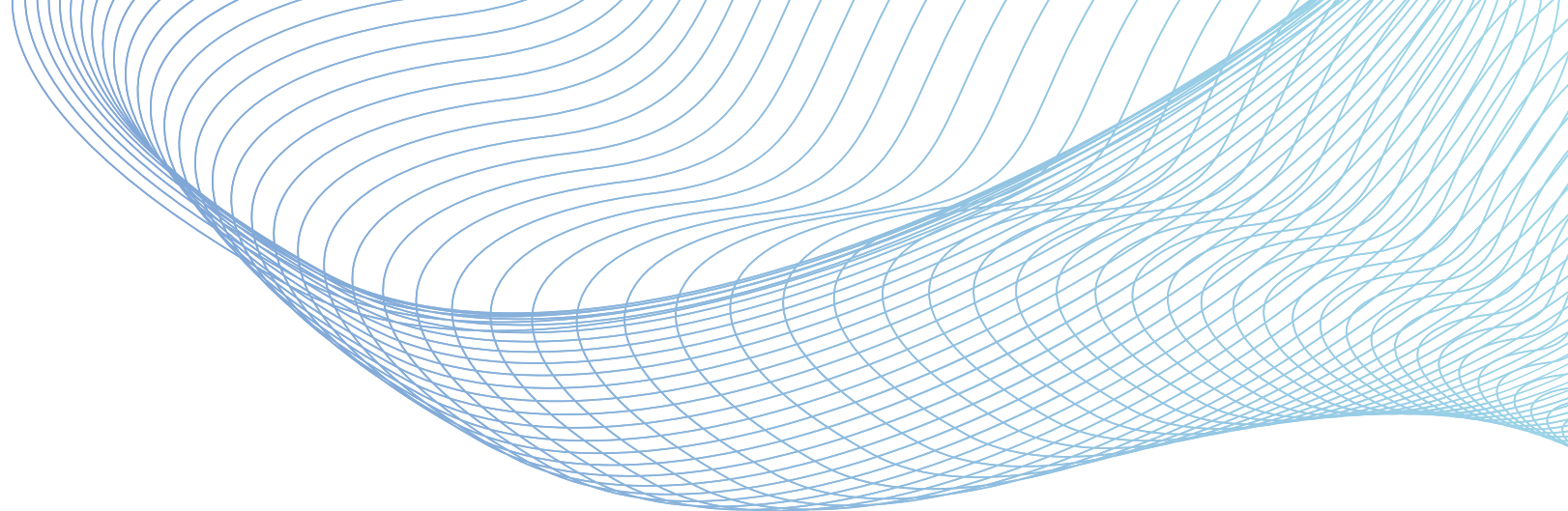
- Variable "Fraud_Found_P" is highly imbalanced

OVER-SAMPLING

- The Synthetic Minority Over-sampling Technique (SMOTE) aims to balance the class distribution by generating synthetic examples from the minority class



OVER-SAMPLING

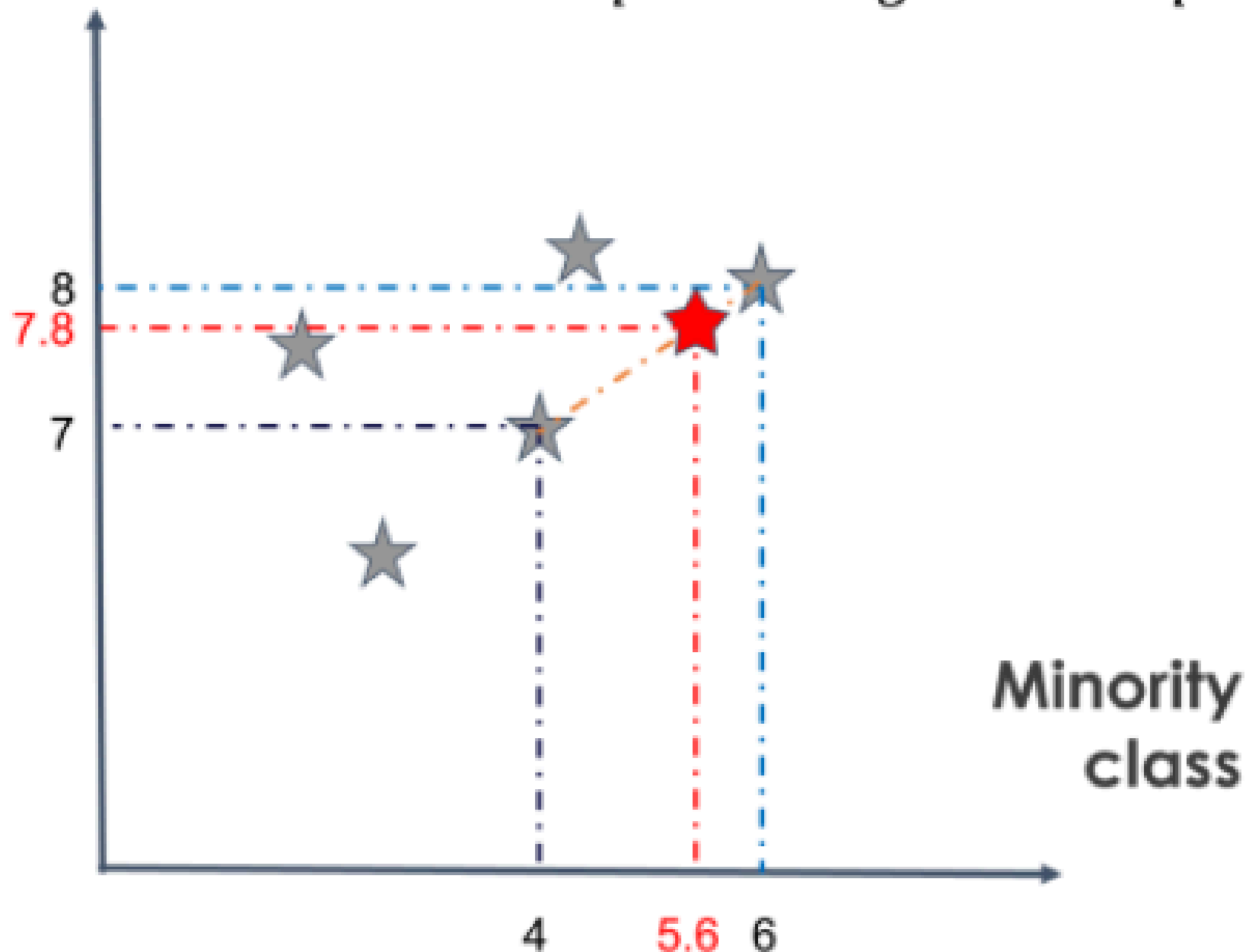


The SMOTE algorithm works in the following way:

- Select a minority class sample from the original dataset.
- Find its k nearest minority class neighbors in the feature space.
- Randomly select one of the k nearest neighbors.
- Generate a new synthetic sample by interpolating between the selected minority class sample and the randomly selected neighbor.
- Repeat steps 1-4 until the desired number of synthetic samples is generated.

SMOTE: numerical example

*New sample = original sample - factor * (original sample - neighbour)*



$$X_{ori} = (4, 7)$$

$$X_{neig} = (6, 8)$$

$$New\ sample = (4, 7) - 0.8 * ((4, 7) - (6, 8))$$

$$New\ sample = (4, 7) - 0.8 * ((-2, -1))$$

$$New\ sample = (4, 7) - ((-1.6, -0.8))$$

EXAMPLES

from imblearn.over_sampling import SMOTE

smote = SMOTE(

sampling_strategy='auto',

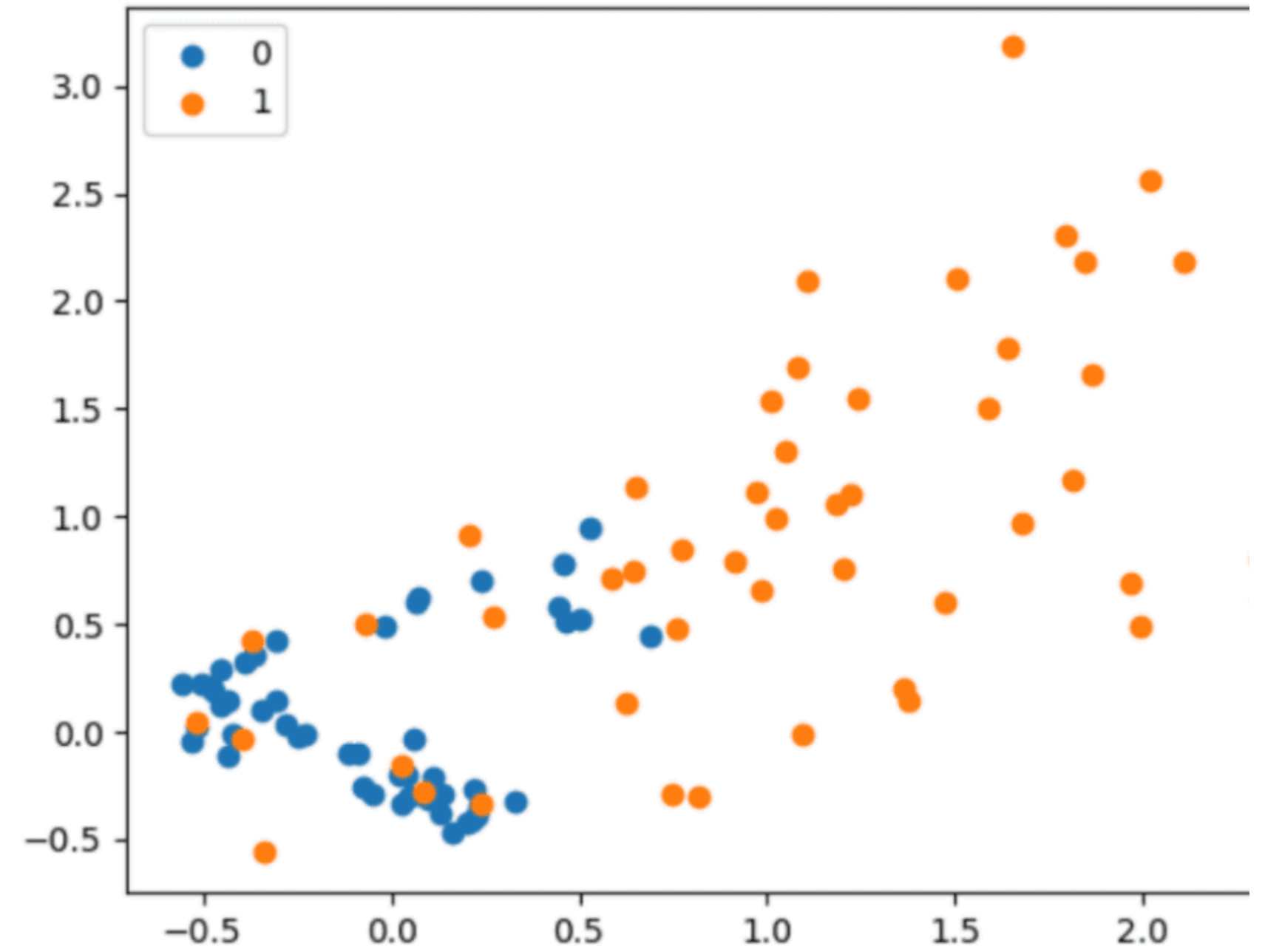
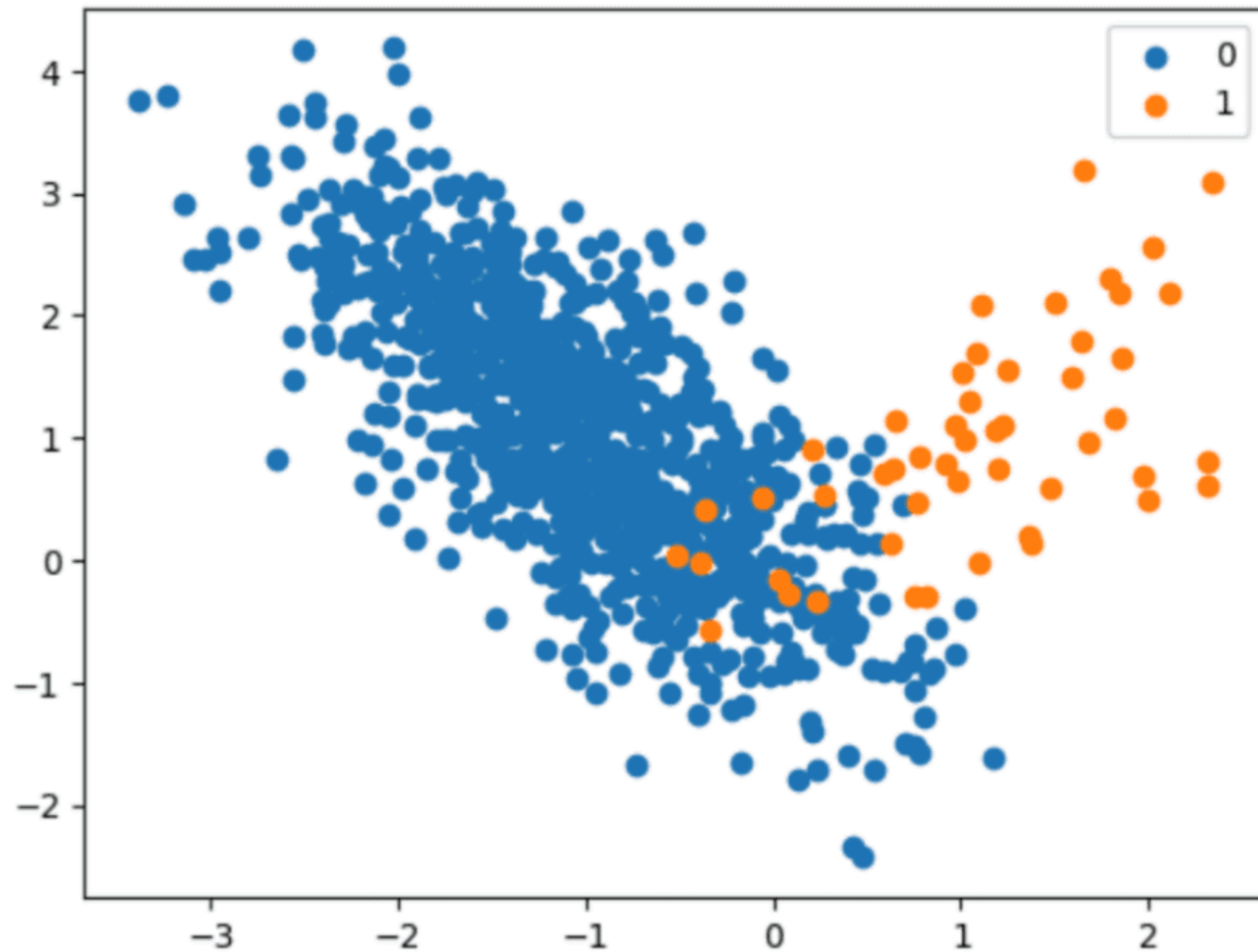
random_state=0,

k_neighbors=5,)

data_res, target_res = smote.fit_resample(data, target)

- The k parameter in SMOTE determines the number of nearest neighbors to consider when generating synthetic samples.

UNDER-SAMPLING

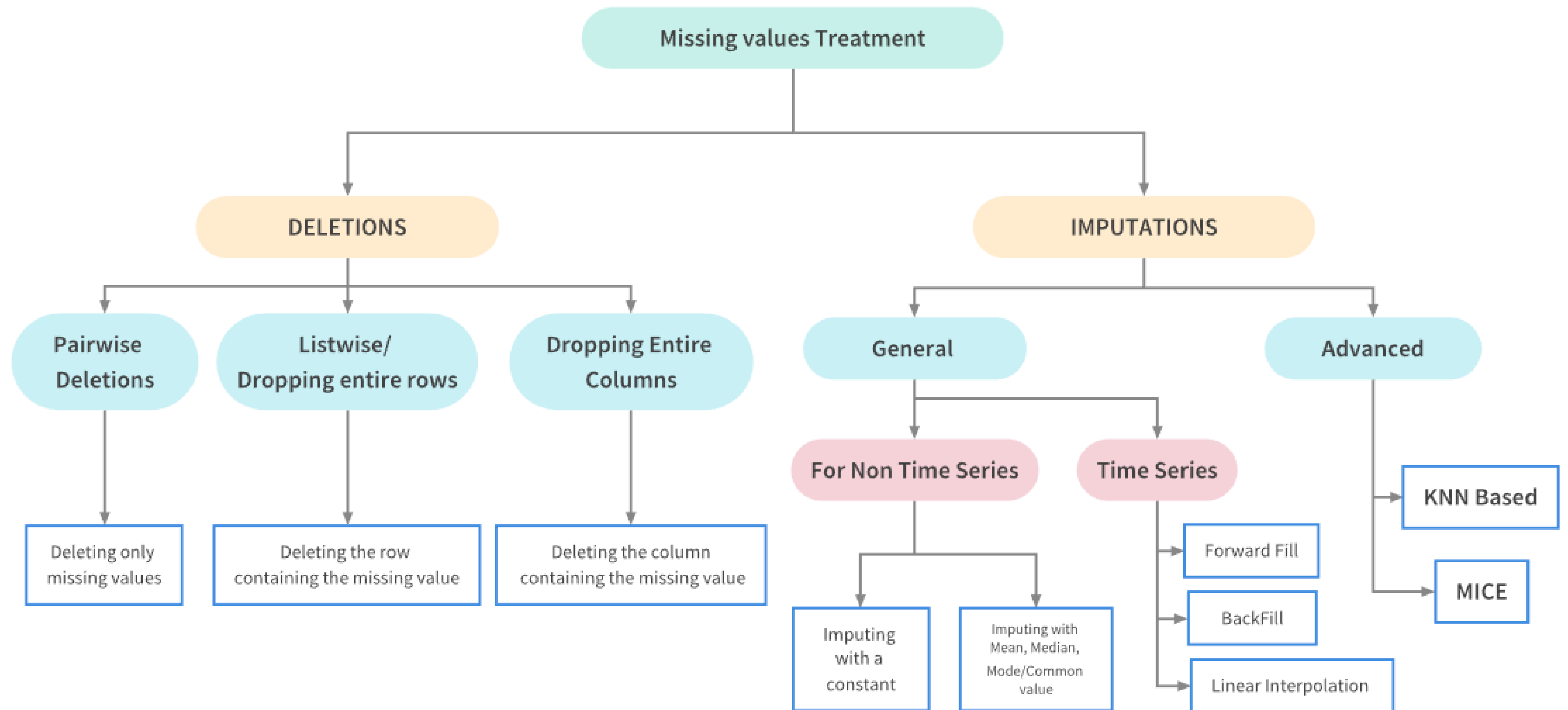


HANDLING MISSING DATA

- Replace the missing value with some constant, specified by the analyst.
- Replace the missing value with the field mean (for numeric variables) or the mode (for categorical variables).
- Replace the missing values with a value generated at random from the observed distribution of the variable.
- Imputation of missing data:
 - (i) continuous variables and (ii) categorical variables.

Mean imputation vs. multiple imputation

HANDLING MISSING DATA



REASONS FOR MISSING VALUES

1. Missing Completely at Random (MCAR)

- The missing values on a given variable (Y) are not associated with other variables in a given data set or with the variable (Y) itself.

2. Missing at Random (MAR)

- MAR occurs when the missingness is not random, but where missingness can be fully accounted for by variables where there is complete information

3. Missing Not at Random (MNAR)

- Missingness depends on unobserved data or the value of the missing data itself.

DELETING MISSING VALUES

- Pairwise Deletion is used when values are missing completely at random i.e MCAR.
- During Listwise deletion, complete rows (which contain the missing values) are deleted. Like Pairwise deletion, listwise deletions are also only used for MCAR values.
- Dropping complete columns: If a column contains a lot of missing values, say more than 80%, and the feature is not significant, you might want to delete that feature. However, again, it is not a good methodology to delete data.

IMPUTATION TECHNIQUES IN PYTHON

Basic Imputations Techniques

- Imputating with a constant value
- Imputation using the statistics (mean, median or most frequent) of each column in which the missing values are located

Imputations Techniques for Time Series Problems

- 'ffill' or 'pad' - Replace NaN s with last observed value
- 'bfill' or 'backfill' - Replace NaN s with next observed value
- Linear interpolation method

```
Date
2015-02-20    7.48
2015-02-21   15.44
2015-02-22    8.47
2015-02-23   28.46
2015-02-24    6.05
2015-02-25    0.81
2015-02-26    NaN
2015-02-27    NaN
2015-02-28    NaN
2015-03-01    1.32
2015-03-02    0.22
2015-03-03    2.25
2015-03-04    1.55
2015-03-05    4.13
Name: Xylene, dtype: float64
```

```
Date
2015-02-20    7.48
2015-02-21   15.44
2015-02-22    8.47
2015-02-23   28.46
2015-02-24    6.05
2015-02-25    0.81
2015-02-26    0.81
2015-02-27    0.81
2015-02-28    0.81
2015-03-01    1.32
2015-03-02    0.22
2015-03-03    2.25
2015-03-04    1.55
2015-03-05    4.13
2015-03-06    4.13
Name: Xylene, dtype: float64
```

IMPUTATION TECHNIQUES IN PYTHON

Linear interpolation method

- an imputation technique that assumes a linear relationship between data points and utilises non-missing values from adjacent data points to compute a value for a missing data point.

`DataFrame.interpolate(method='linear', *, axis=0, limit=None, inplace=False, limit_direction=None, limit_area=None, downcast=_NoDefault.no_default, **kwargs)`

- **inplace**: bool, default False

Update the data in place if possible.

Advanced Imputation Techniques

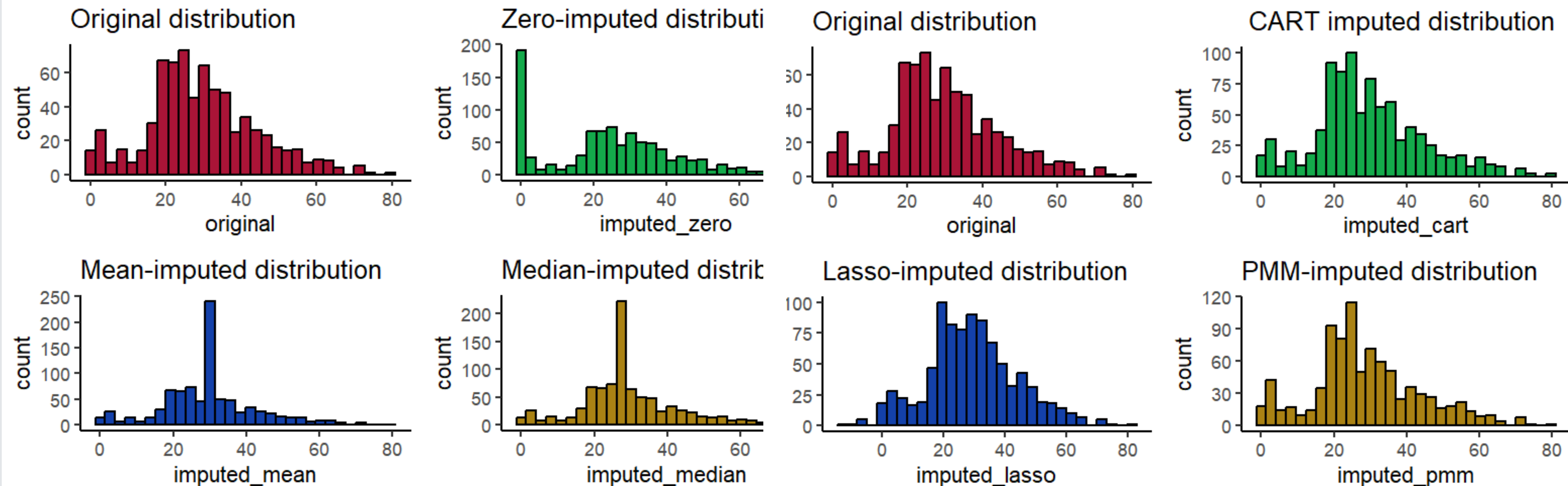
- **K-Nearest Neighbor Imputation**
- **Multivariate imputation by chained equations (MICE)**: A strategy for imputing missing values by modeling each feature with missing values as a function of other features in a round-robin fashion. It performs multiple regressions over random sample of the data, then takes the average of the multiple regression values and uses that value to impute the missing value

Some algorithms like XGBoost and LightGBM can handle missing values without any preprocessing, by supplying relevant parameters

IMPUTATION TECHNIQUES IN R ON TITANIC

Feature: Age

library(mice)



IMPUTATION TECHNIQUES IN R ON TITANIC

```
library(mice)
```

```
titanic_numeric <- titanic_train %>%
```

```
  select(Survived, Pclass, SibSp, Parch, Age)
```

```
mice_imputed <- data.frame(
```

```
  original = titanic_train$Age,
```

```
  imputed_pmm = complete(mice(titanic_numeric, method = "pmm"))$Age,
```

```
  imputed_cart = complete(mice(titanic_numeric, method = "cart"))$Age,
```

```
  imputed_lasso = complete(mice(titanic_numeric, method = "lasso.norm"))
```

```
$Age )
```

HOMework

- Apply all these data exploration & data preparation techniques to another dataset using **R/Python**

THANK YOU

